

1 v0.8 치수·검사 조립 도면 세트

물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact donor 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

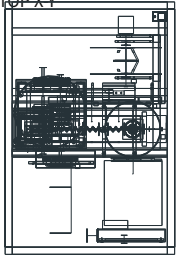
Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

2 GA-001 — general arrangement

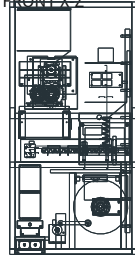
GA-001 general arrangement

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 187

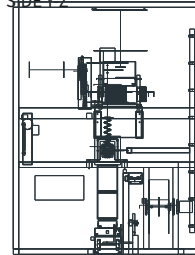
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선

Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

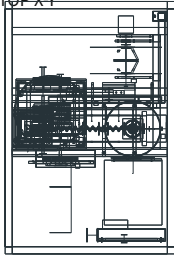
재료/구성: mixed assembly; see BOM.csv 핵심 치수/공차: 470 × 700 × 930 mm envelope; service aisle ≥ 600 mm 단위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

3 ASM-001 — full assembly

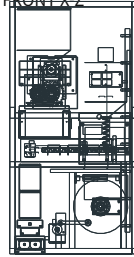
ASM-001 full assembly

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 187

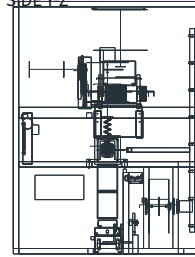
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

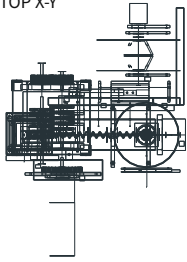
재료/구성: mixed assembly; see BOM.csv **핵심 치수/공차:** high-load path cutter/screw → metal bearing plate → profile → table **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

4 ASM-002 — module arrangement

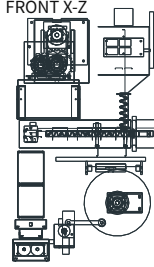
ASM-002 module arrangement

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 137

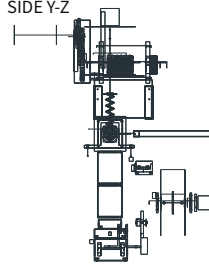
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 427.00 · Y 601.00 · Z 760.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

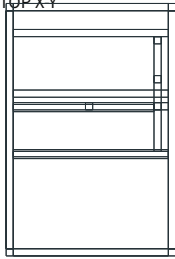
재료/구성: mixed assembly; see BOM.csv **핵심 치수/공차:** module datum transfer ≤ 0.50 mm; service modules removable without hot-path disturbance **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

5 FR-001 — frame

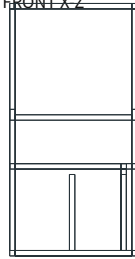
FR-001 frame

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 28

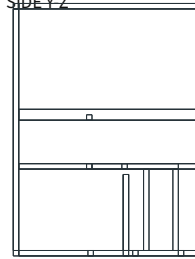
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

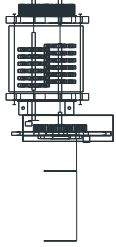
재료/구성: 2020/2040 aluminum profile **핵심 치수/공차:** base 470 × 700 mm; anchor M8 × 4; rail squareness ≤ 0.50/700
단위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

6 SH-001 — shredder assembly

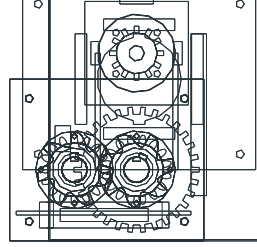
SH-001 shredder assembly

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 40

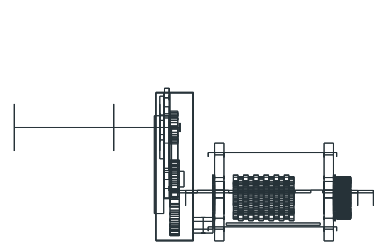
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 195.00 · Y 459.00 · Z 195.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

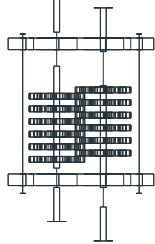
재료/구성: steel cutter module **핵심 치수/공차:** shaft centres 48.00 ± 0.03 mm; rotating-to-static clearance ≥ 1.90 mm **단**
위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

7 SH-002 — cutter stack

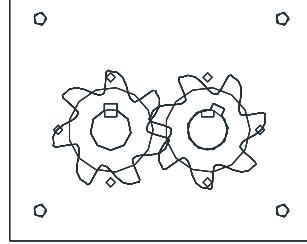
SH-002 cutter stack

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 20

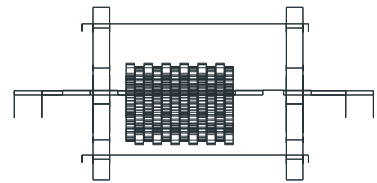
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 150.00 · Y 260.00 · Z 125.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

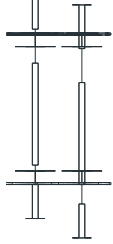
재료/구성: D2 cutters / steel spacers **핵심 치수/공차:** CUT-01 t6 and CUT-02 t7; axial gap 0.25–0.50 mm by metal shim **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

8 SH-003 — shaft and bearing assembly

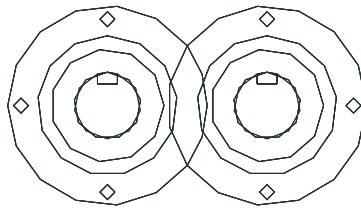
SH-003 shaft bearing

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 8

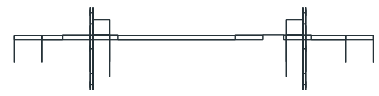
TOP X-Y



FRONT X-Z



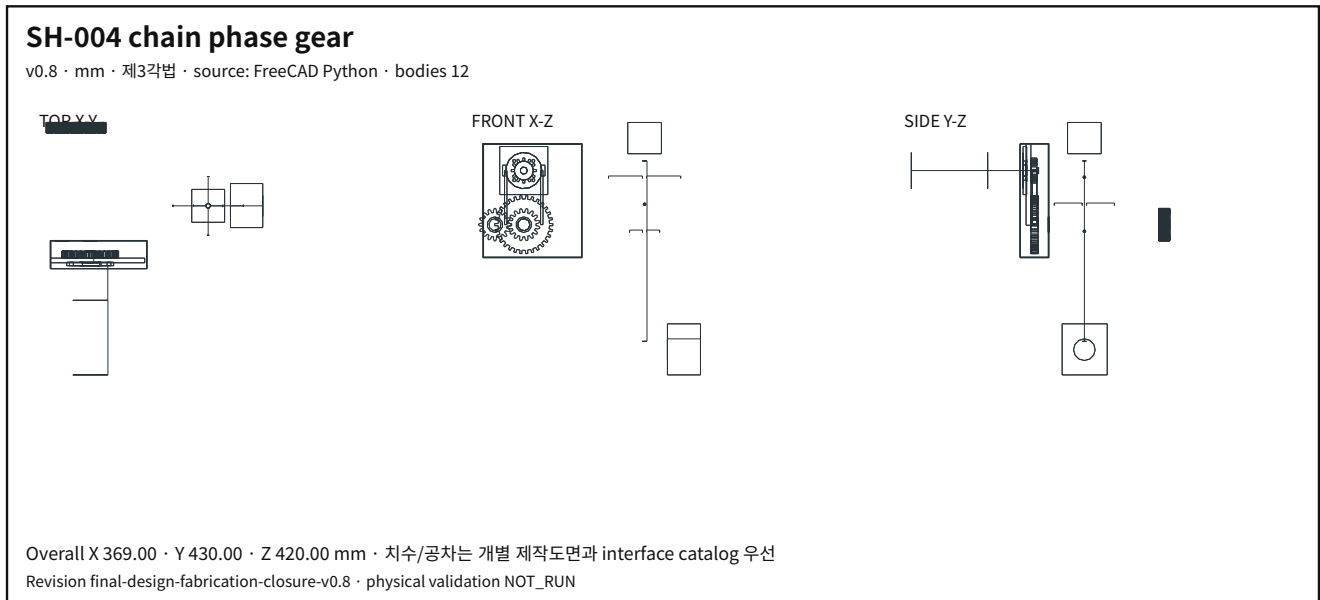
SIDE Y-Z



Overall X 108.00 · Y 260.00 · Z 60.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

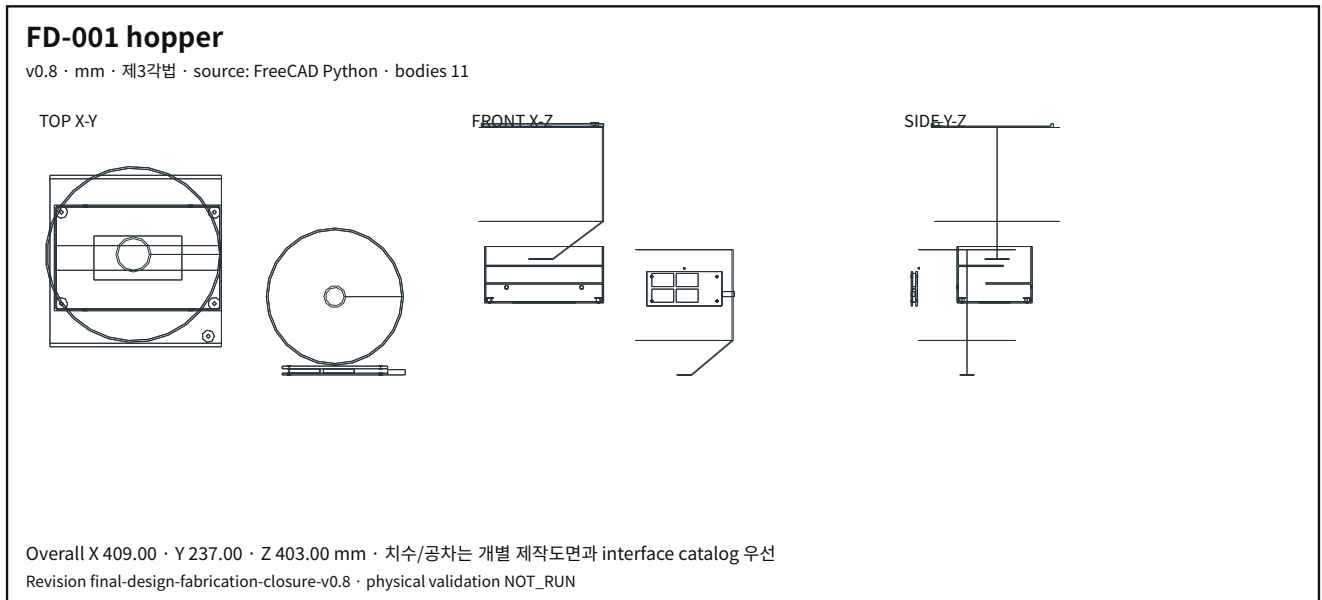
재료/구성: S45C shafts / 6004-2RS 핵심 치수/공차: Ø20 h6 seats; shaft TIR ≤ 0.05 mm; centre parallelism $\leq 0.10/150$ 단위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

9 SH-004 — chain and phase gear



재료/구성: S45C keyed hubs/gears / 35 chain **핵심 치수/공차:** 12T:30T; chain alignment ≤ 0.50 mm; midspan slack 2–3%
단위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

10 FD-001 — hopper



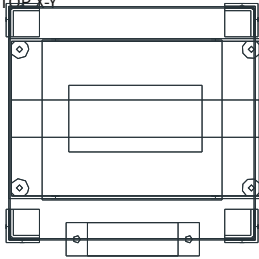
재료/구성: 5052-H32 hopper **핵심 치수/공차:** feed opening 150×150 mm; all reachable edges $R/C \geq 0.5$ mm **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

11 FD-002 — recirculation/screen

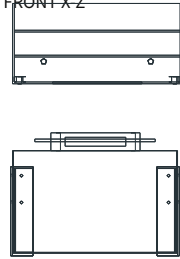
FD-002 recirculation screen

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 8

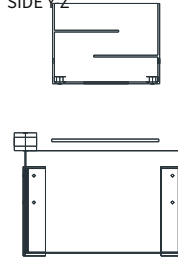
TOP X-Y



FRONT X-Z



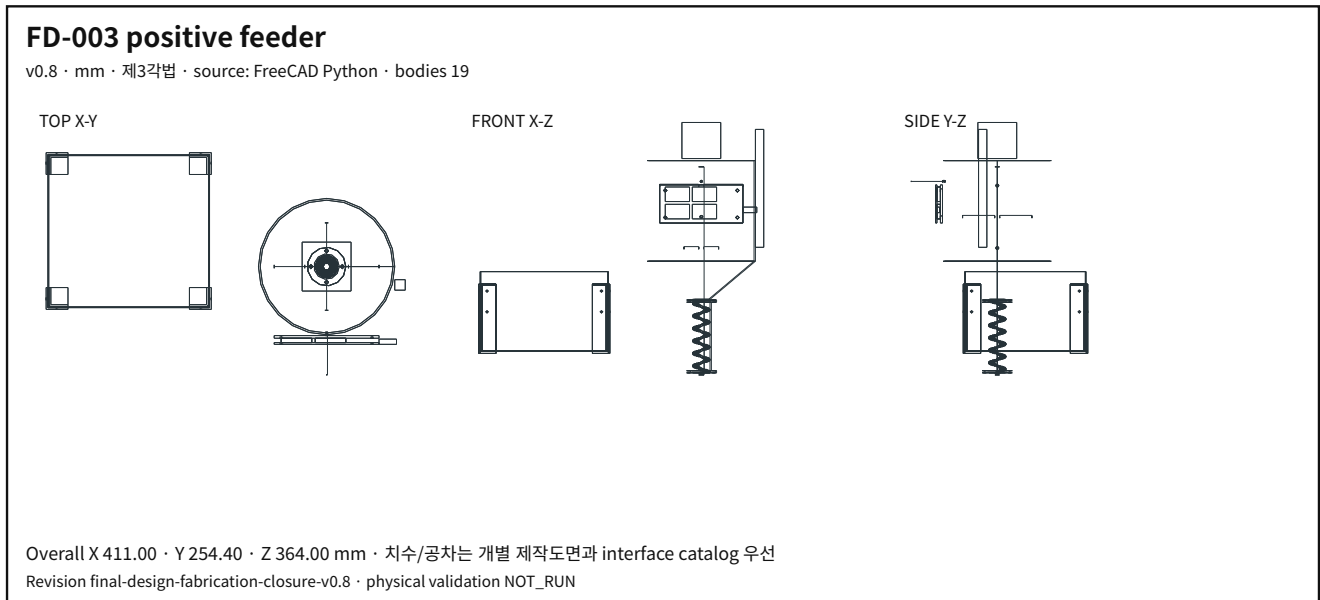
SIDE Y-Z



Overall X 192.00 · Y 189.40 · Z 283.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: 304 screen / sheet chute **핵심 치수/공차:** screen aperture Ø5 on 9 pitch; cutter/static clearance ≥ 1.90 mm 단
위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

12 FD-003 — positive feeder



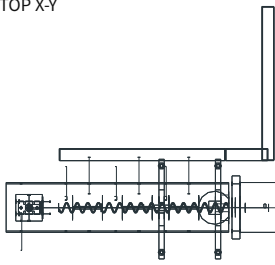
재료/구성: 304 auger/housing/common agitator shaft **핵심 치수/공차:** auger OD24.60; housing ID25.00 +0.05/0; radial clearance 0.20–0.25 mm; pitch18 **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

13 EX-001 — extruder assembly

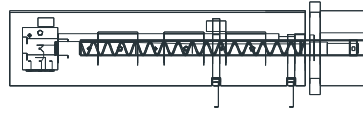
EX-001 extruder assembly

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 33

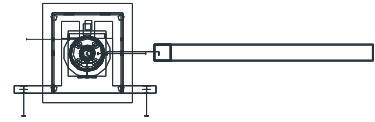
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 407.00 · Y 380.00 · Z 119.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: SCM440 screw/barrel / steel supports **핵심 치수/공차:** rear axial datum fixed; front guide axial travel ≥ 1.30 mm **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

14 EX-002 — screw/barrel/die

EX-002 screw barrel die

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 13

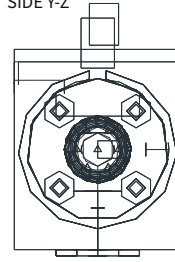
TOP X-Y



FRONT X-Z



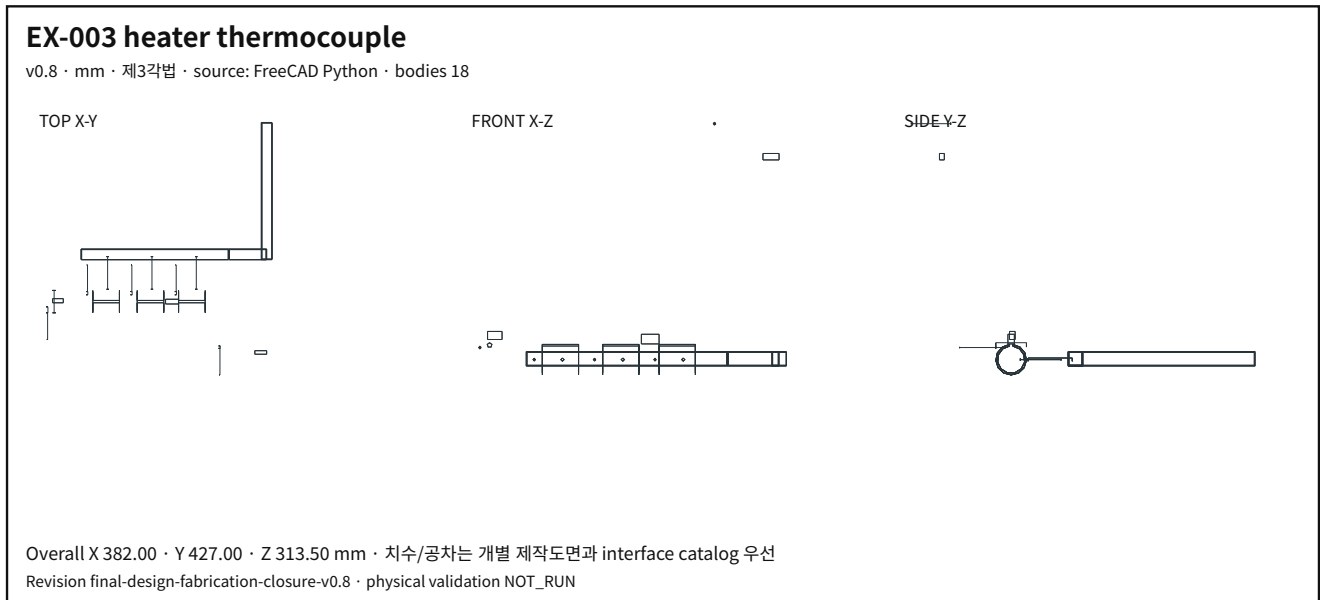
SIDE Y-Z



Overall X 380.50 · Y 40.00 · Z 60.50 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

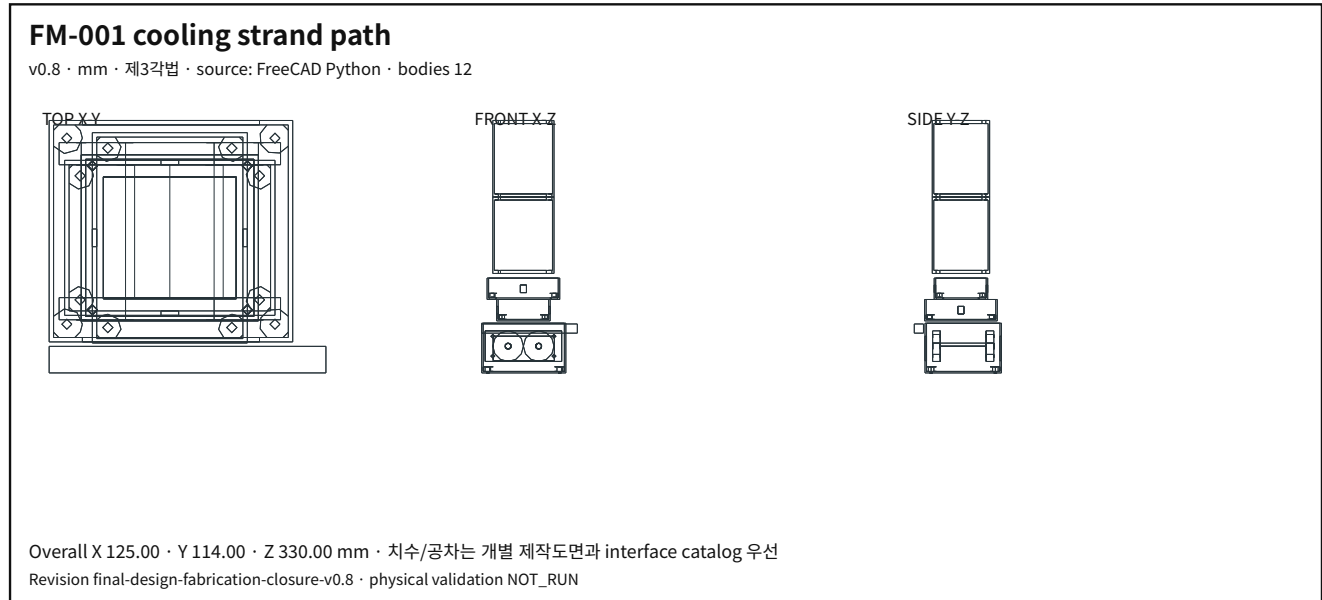
재료/구성: nitrided SCM440 / 17-4PH die insert **핵심 치수/공차:** cold diametral clearance 0.28–0.32 mm; coaxiality ≤ 0.05 mm **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

15 EX-003 — heater/thermocouple layout



재료/구성: mica/NiCr heater and MI thermocouple **핵심 치수/공차:** probe insertion ≥ 12 mm; heater-to-polymer path metal-only; shield clearance ≥ 12 mm **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

16 FM-001 — cooling and strand path



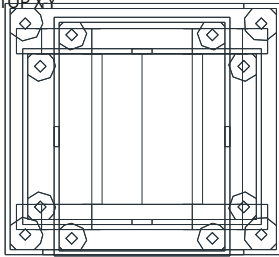
재료/구성: 5052 duct / donor fans **핵심 치수/공차:** strand centreline offset ≤ 0.50 mm; hot-shield clearance ≥ 12 mm 단
위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

17 FM-002 — gauge/puller

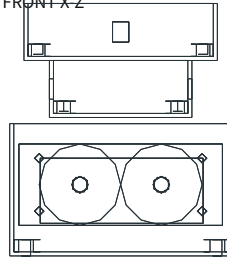
FM-002 gauge puller

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 9

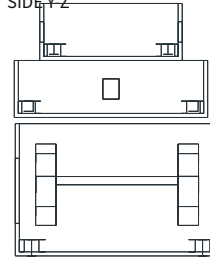
TOP X-Y



FRONT X-Z



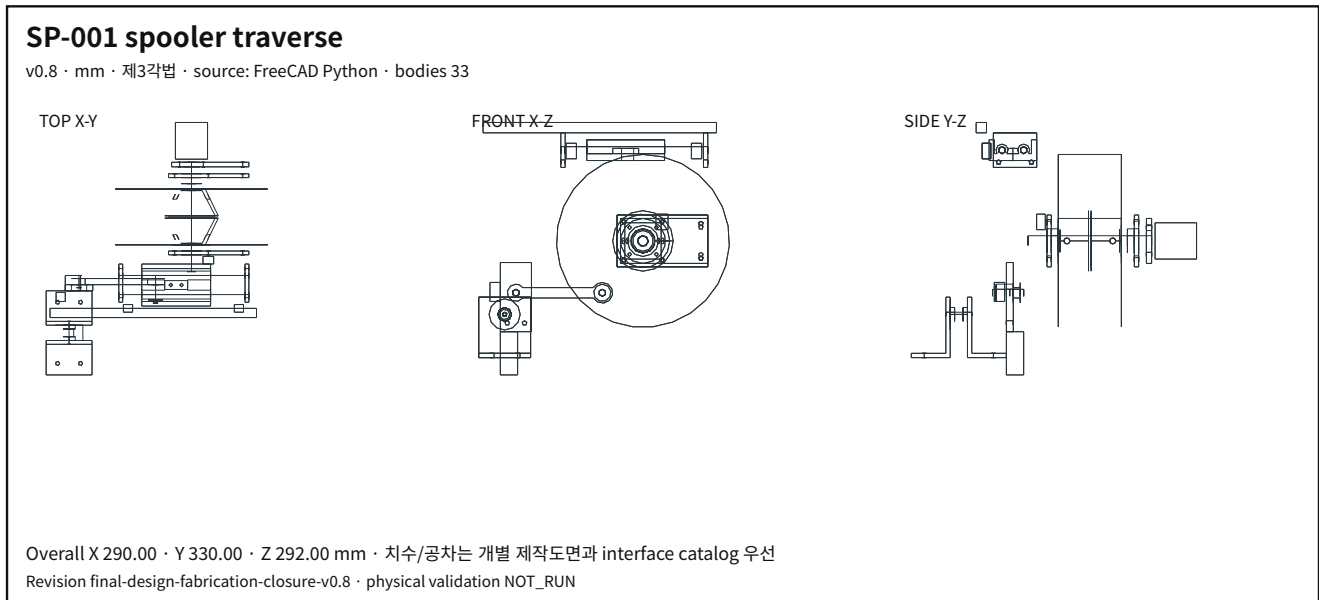
SIDE Y-Z



Overall X 110.00 · Y 100.50 · Z 124.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

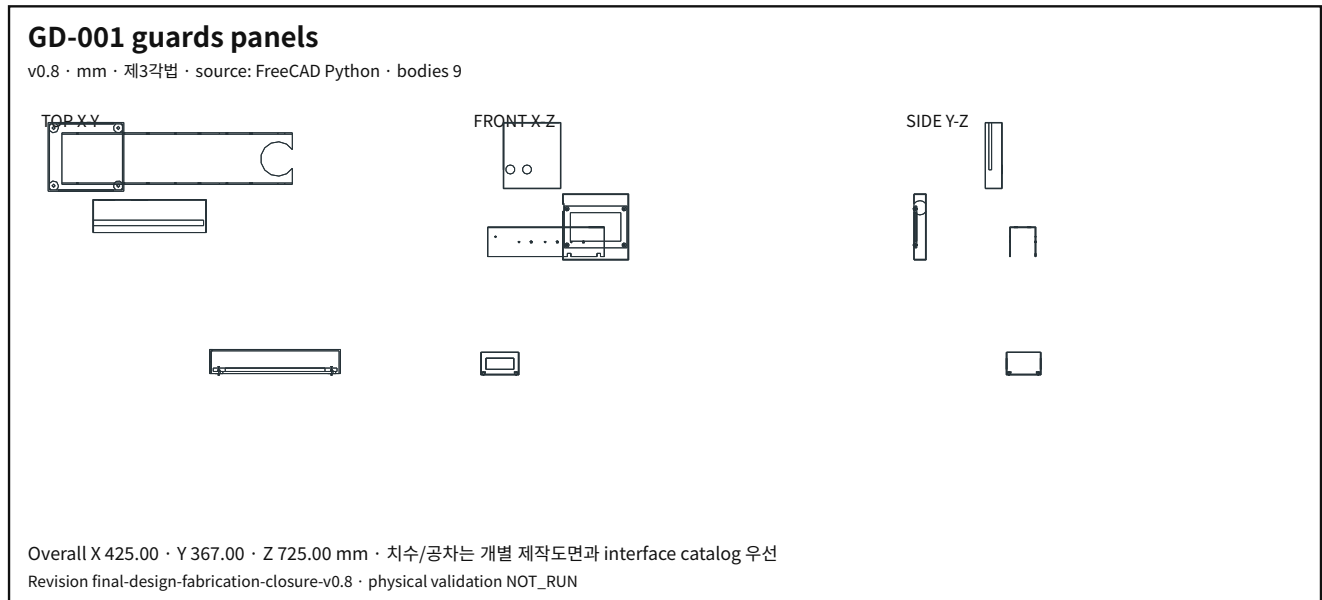
재료/구성: 6061 plates / POM-C rollers 핵심 치수/공차: roller axes parallel $\leq 0.05/80$; gauge datum alignment ≤ 0.10 mm
단위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

18 SP-001 — spooler/traverse



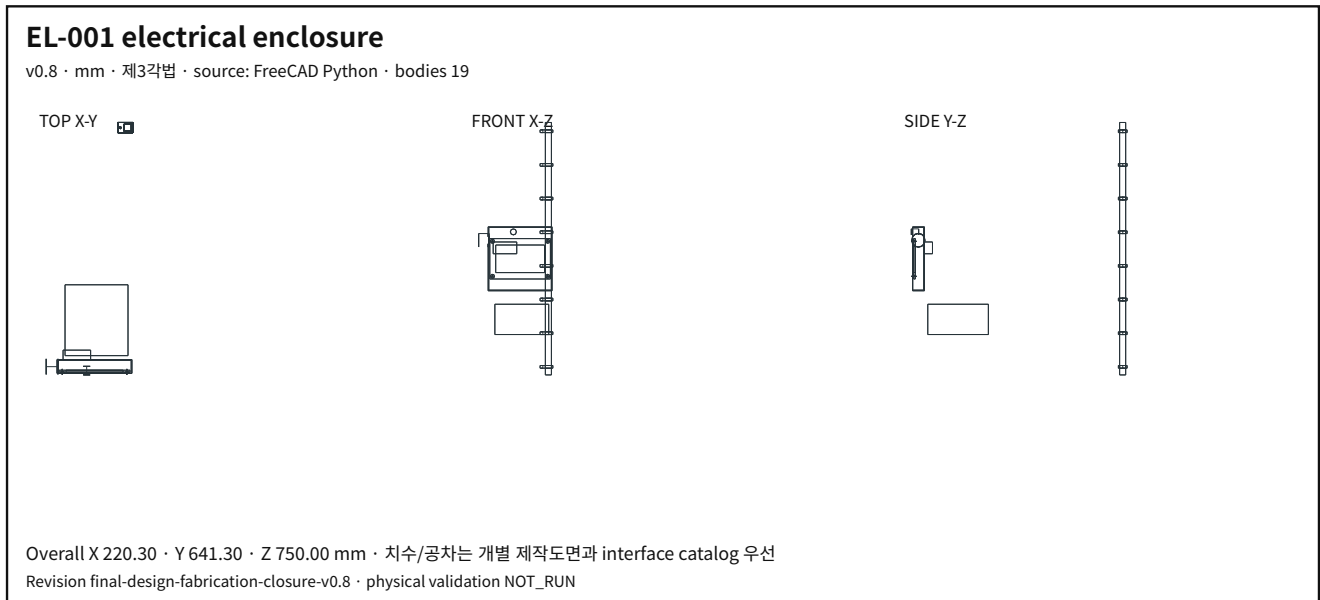
재료/구성: 6061 plates / stainless shafts **핵심 치수/공차:** spool shaft $\varnothing 12$ h6; traverse rod parallelism $\leq 0.10/160$ **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

19 GD-001 — guards and panels



재료/구성: polycarbonate and bonded metal panels **핵심 치수/공차:** hazard opening ≤ 6 mm; no reach path to moving/hot parts **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

20 EL-001 — electrical enclosure



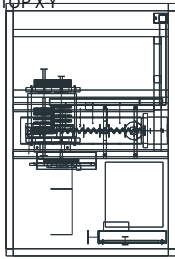
재료/구성: 2 mm 5052 enclosure **핵심 치수/공차:** PE bond target 0.10 ohm 이하; signal/power duct separation ≥ 18 mm
단위: mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35

21 SV-001 — service envelopes

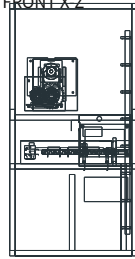
SV-001 service envelopes

v0.8 · mm · 제3각법 · source: FreeCAD Python · bodies 120

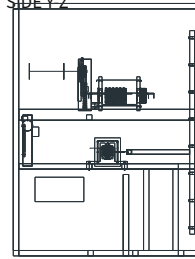
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: service-envelope reference geometry **핵심 치수/공차:** front/rear access ≥ 600 mm; hot-zone removal envelope kept clear **단위:** mm · 제3각법 · NTS · source adc8bec89d946f696c179d79b1af2c401189ab35